



An AI-Powered Digital Twin for Predictive Monitoring and Decision Support in Construction Project Management

Nada Hammad Aljohani¹

¹ King Abdulaziz University
Faculty of Computing and Information Technology
Professional Master in Artificial Intelligence

Supervised by:

Dr. Somayah Albaradei

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Abstract—Construction projects in Saudi Arabia, particularly Vision 2030 megaprojects, consistently suffer from schedule overruns, cost escalations, and fragmented information systems — a challenge underscored by the fact that over 70% of major construction projects globally are delivered late or over budget [1]. This research proposes an AI-Powered Digital Twin for construction project management that operates on schedule level data rather than physical sensor inputs, addressing these challenges through four integrated modules: an AI Delay Predictor using XGBoost and Random Forest classifiers with SHAP explainability; a What If CPM Simulator for real time schedule scenario analysis; a Proactive Risk Alert Engine that monitors key performance indicators and surfaces early warnings before delays escalate; and a Generative AI Advisory Chat that enables project managers to receive contextual recovery recommendations through natural language interaction. Unlike existing Digital Twin frameworks that rely on IoT sensors or BIM infrastructure, the proposed platform operates on standard Primavera P6 schedule data, making it directly deployable within existing construction project management workflows. The system will be trained on a synthetic Primavera P6 dataset of 10,000 activities calibrated from a real Saudi Arabian construction megaproject and validated at both activity and project levels. Performance targets are set at AUC-ROC 0.80 and F1-Score 0.75, consistent with benchmarks established in the construction delay prediction literature. Collectively, the four modules are designed to transform the Digital Twin from a passive data mirror into a proactive, conversational decision support system. **Keywords:** Digital Twin, Construction Delay Prediction, XGBoost, SHAP, Critical Path Method, Generative AI, Saudi Arabia, Vision 2030.

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I. INTRODUCTION

a. Problem Definition and Background

Construction project management is among the most data intensive and risk prone disciplines in the built environment. Large scale projects encompass hundreds of

interdependent work packages, multi million riyal budgets, and complex networks of contractors, suppliers, and regulatory bodies. Despite widespread adoption of enterprise project management platforms such as Primavera P6 and Oracle Unifier, project managers continue to depend on manual processes for progress monitoring, risk identification, and change impact analysis practices that systematically delay problem detection and constrain effective decision making. Research consistently reports that more than 70% of major construction projects are delivered late or over budget, with root causes frequently

attributed to late risk identification, poor inter task visibility, and the absence of predictive analytical capability (Egwim et al., 2021).

In the Kingdom of Saudi Arabia, this challenge is compounded by the concurrent execution of hundreds of Vision 2030 megaprojects, which demand sophisticated project oversight and data driven governance at an unprecedented scale. Three persistent operational failures characterize current practice in Saudi Arabian construction project management: slow manual reporting that renders progress data obsolete before it reaches decision makers; fragmented information systems in which schedule, cost, document workflow, and progress data reside across multiple disconnected platforms; and intuition based decision making in the absence of predictive analytics, whereby project managers assess delay risk and evaluate recovery options without quantitative decision support.

The motivation for this research derives from direct professional experience within the construction project management sector. In the capacity of PMIS Manager on a large scale Saudi Arabian construction megaproject, the researcher observed these systemic failures at first hand. Progress reporting routinely required multiple days of manual data consolidation from Primavera P6, Oracle Unifier, and numerous supplementary workbooks, with the resultant information frequently rendered obsolete by the time it reached senior stakeholders. Risk assessment in project meetings was characterized by experience based judgement rather than analytical evidence, and the evaluation of schedule recovery strategies necessitated hours of manual recalculation — further compressing the decision window available to project teams. Contractors, consultants, and client representatives shared a common operational frustration: despite the abundance of project data distributed across organizational systems, no integrated intelligence layer existed to transform that data into timely, actionable insight. This research constitutes a direct and systematic response to those observed deficiencies, grounded in the conviction that the convergence of Digital Twin technology and Artificial Intelligence can fundamentally transform the practice of construction project management in the Saudi Arabian context.

b. Project Objectives and Scope

This study presents an AI-Powered Digital Twin platform for construction project management, defined as a continuously updated virtual representation of a project's operational state, designed to address the three identified failures through four integrated intelligent modules. The AI Delay Predictor employs XGBoost and Random Forest classifiers to forecast activity level delay probability, with SHAP (SHapley Additive exPlanations) values providing transparent, feature level explain ability for every prediction. The What If CPM Simulator enables real time modelling of schedule change scenarios and quantification of their downstream impact on project completion dates and estimated costs. The Proactive Risk Alert Engine continuously monitors key performance indicators, including the Schedule Performance Index, document review cycle duration, and submittal rejection rates,

surfacing early warning signals before delays escalate beyond the point of efficient recovery. The Generative AI Advisory Chat facilitates natural language interaction, enabling project managers to receive contextual recovery recommendations, schedule acceleration strategies, and resource reallocation proposals in response to emerging challenges.

The scope of this study is delineated by three parameters. First, data input is facilitated through an Excel based upload of Primavera P6 schedule data, without requiring IoT sensor infrastructure or Building Information Modelling integration. Second, the predictive models were trained on a synthetic dataset of 10,000 Primavera P6 activities calibrated from the statistical properties of a real Saudi Arabian construction megaproject, ensuring domain realism while preserving project confidentiality. Third, the platform is deployed as a Streamlit web application, accessible to non technical PMO staff without the need for software installation.

c. Relevance and Significance

The significance of this study extends across three complementary dimensions. From an academic perspective, it addresses a documented gap in the Digital Twin literature: no existing study has demonstrated a deployable, AI-powered Digital Twin constructed using Primavera P6 structured data, validated against Saudi Arabian megaproject statistics, and incorporating both explainable delay prediction and generative AI advisory capability within a single integrated platform. The two level validation strategy, which combines activity level model evaluation with project level generalization assessment, constitutes a methodological contribution to the emerging field of construction artificial intelligence.

From an industry perspective, the platform provides construction firms, PMO departments, and planning engineers with a directly applicable decision support instrument that augments rather than displaces the existing project management infrastructure. By transforming delayed, fragmented reporting into real time predictive intelligence, the system enables earlier risk identification, more rapid response, and more informed resource allocation, addressing the operational deficiencies experienced by project teams on active megaprojects.

At the national level, this research aligns with Saudi Arabia's Vision 2030 digital transformation objectives. As the Kingdom concurrently delivers an unprecedented volume of infrastructure megaprojects, the adoption of AI-powered project intelligence tools represents a scalable and immediately deployable mechanism for elevating construction sector performance and supporting the timely delivery of strategically critical national development.

II. LITERATURE REVIEW

The literature relevant to this research spans three intersecting domains: Digital Twin technology in construction, machine learning for construction delay prediction, and generative AI for decision support in project management.

a. Digital Twin Technology in Construction

Digital Twin technology has attracted substantial and growing research attention in the construction sector. Opoku et al. [2] conducted a foundational literature review establishing the core taxonomy of Digital Twin applications in construction, encompassing structural health monitoring, progress tracking, and lifecycle management. Building upon this foundation, Tuhaise et al. [3]) systematically catalogued the enabling technologies for construction Digital Twins, including Internet of Things devices, cloud computing platforms, and artificial intelligence, in a study published in *Automation in Construction*, one of the field's leading journals. A comprehensive review published by Baghdadi [4], synthesizing 117 peer reviewed papers from 2024 to 2025, confirmed that the integration of Digital Twin technology with artificial intelligence for construction project management represents the dominant emerging research theme in the discipline.

A five level Digital Twin maturity framework, proposed in the literature, categorizes implementations from basic Building Information Modelling visualization at Level 1 through IoT enabled real time monitoring at Level 3 to fully autonomous AI driven control at Level 5. Most current real world implementations remain at Levels 2 and 3, underscoring the substantial research opportunity associated with AI integration at higher maturity levels. Pan and Zhang [5] contributed one of the most widely cited frameworks in this domain, proposing a BIM data mining integrated Digital Twin for advanced project management published in *Automation in Construction*. Crucially, their work demonstrated that structured project management data, rather than IoT sensor streams, can constitute a viable Digital Twin input, providing the conceptual foundation for the data driven approach adopted in this study. A study specific to the Saudi Arabian context, published in Alnaser and Elmousalami [6] surveyed 106, architecture, engineering, and construction firms and identified data interoperability challenges, the absence of standardized implementation frameworks, and a shortage of qualified personnel as the primary barriers to Digital Twin adoption in the Kingdom.

b. Machine Learning for Construction Delay Prediction

The application of machine learning to construction delay prediction has matured considerably over the past five years. Egwim et al. [1] evaluated twenty seven machine learning models for delay prediction using real project data, establishing that ensemble methods — particularly XGBoost and Random Forest — consistently outperform simpler classifiers due to their capacity to model complex, non-linear relationships within construction project datasets. A 2025 study published in the *Journal of Mechanics of Continua and Mathematical Sciences* corroborated this finding, confirming that gradient-boosted ensemble methods exhibit superior accuracy and predictive capacity for the classification of construction delays. Of particular relevance to the present research, Alsulamy [7] benchmarked CatBoost, XGBoost, and LightGBM specifically on Saudi

Arabian construction project datasets, finding that gradient-boosted ensemble methods — including LGBM and XGBoost — consistently deliver the strongest delay prediction performance in the Saudi context. XGBoost is adopted as the primary classifier in this study on the basis of its superior compatibility with SHAP explainability and its demonstrated performance on activity-level tabular data in the broader construction delay prediction literature. The integration of SHAP (SHapley Additive exPlanations) into construction delay prediction models has been identified as a critical advancement for practitioner adoption. By providing transparent, feature-level explanations for individual predictions, SHAP transforms opaque ensemble classifiers into interpretable decision-support instruments that project managers can interrogate and trust [8].

c. Generative AI for Project Decision Support

The application of large language models to construction project management represents an emerging and largely unexplored research area. A 2026 peer reviewed study from the University of East London, published in *Frontiers in Built Environment*, confirmed that current construction management tools, including scheduling platforms, safety monitoring systems, and risk registers, operate in isolation, and that linking AI driven risk prediction to automated advisory response constitutes the most significant unresolved gap in construction digital intelligence [9]. While generative AI has been applied to project management in adjacent domains, its integration within a construction specific Digital Twin framework — enabling natural language interaction between project managers and an AI advisory system — has not been demonstrated in the published literature.

d. Research Gaps

A systematic analysis of the literature across all three domains reveals five critical gaps that this research is uniquely positioned to address.

TABLE 1: SUMMARY OF RESEARCH GAPS, EXISTING LITERATURE, AND STUDY CONTRIBUTIONS

Research Gap	State of Existing Literature
No integrated AI-powered Digital Twin platform for construction project management	Digital Twin and AI are studied in isolation or applied to narrow subdomains [3]
Absence of Saudi specific construction delay prediction models	Existing ML delay studies rely predominantly on international and non Saudi datasets [1])
No CPM based what if simulation embedded within a Digital Twin framework	Schedule simulation tools operate independently of Digital Twin environments [5]
No generative AI advisory layer within a construction Digital Twin	Generative AI has not been applied to construction specific Digital Twin decision support [9]
No proactive risk alerting based on continuous KPI monitoring	Existing systems report on delays after they occur rather than predicting them in advance

The convergence of Saudi Arabia’s Vision 2030 construction ambitions, the demonstrated superiority of XGBoost and Random Forest for local delay prediction, and the complete absence of any integrated deployable platform addressing all five gaps collectively establish a compelling and timely research opportunity. This study represents the first attempt to unify explainable machine learning, Critical Path Method simulation, proactive risk alerting, and generative AI advisory capability within a single construction project management Digital Twin, validated against real Saudi Arabian megaproject data.

III. METHODOLOGY

a. Chosen Methods, Tools, and Techniques

This study adopts Design Science Research (DSR) as its overarching methodological framework, a well established paradigm in information systems and engineering research that emphasizes the construction and empirical evaluation of artifacts as the primary mode of knowledge contribution [10].

The research is structured into two sequential phases: (i) artifact construction, which encompasses system design and implementation, and (ii) empirical evaluation, which includes accuracy benchmarking and two level model validation. The overall methodological workflow is illustrated in Fig. 1.

b. Predictive Modelling — XGBoost and Random Forest

Two ensemble classifiers are employed as the primary predictive components of the AI Delay Predictor module. These models are selected based on their demonstrated

superiority in construction delay prediction literature and their empirical validation in the Saudi Arabian context.

XGBoost (Extreme Gradient Boosting) serves as the primary classifier due to its superior performance on structured tabular data, which aligns with the nature of construction project datasets derived from scheduling and project management systems. The model is particularly well suited for handling non linear relationships and heterogeneous feature distributions commonly observed in construction data. In addition, XGBoost natively addresses class imbalance through the `scale_pos_weight` parameter, which is critical for delay prediction tasks where delayed instances are typically underrepresented.

Furthermore, XGBoost is fully compatible with SHAP based explainability, enabling transparent interpretation of model predictions, which is essential for decision support in project management environments. Its effectiveness has also been empirically validated in Saudi Arabian construction contexts [7]. The model is configured with 300 estimators, a maximum tree depth of 6, a learning rate of 0.05, and subsampling ratios of 0.8 for both observations and features.

Random Forest is employed as a benchmark classifier to provide a robust and interpretable baseline for comparison. As an ensemble of decision trees trained via bootstrap aggregation, it is inherently resistant to noise and overfitting, making it suitable for complex and imperfect real world datasets. Random Forest is widely adopted in construction delay prediction literature, which supports its inclusion for comparative evaluation. The model utilizes balanced class weighting to address class imbalance and is configured with 300 trees and a maximum depth of 12.

The selection of these two tree based ensemble models is further justified by their demonstrated superiority over deep learning approaches in tabular data contexts, where neural networks often fail to outperform ensemble methods without significantly larger datasets.

Both models are evaluated using Precision, Recall, F1-score, and AUC-ROC as primary metrics. Particular emphasis is placed on recall to minimize false negatives (i.e., undetected delays), which carry the highest operational risk in construction project management. Performance targets are defined as F1-score ≥ 0.75 and AUC-ROC ≥ 0.80 , consistent with benchmarks reported in the literature.

c. Explainability — SHAP

To enhance model transparency and support informed decision making, SHapley Additive exPlanations (SHAP) values are computed post training using the TreeExplainer algorithm for both classifiers. SHAP enables multi level interpretability of model behavior. At the global level, feature importance is quantified through the aggregation of mean absolute SHAP values across all predictions, allowing the identification of the most influential factors contributing to delay prediction. At the feature level, the direction and magnitude of feature effects are examined using beeswarm visualizations, which illustrate how variations in feature values influence the predicted probability of delay. At the individual level, prediction specific explanations are generated using waterfall plots, which decompose each

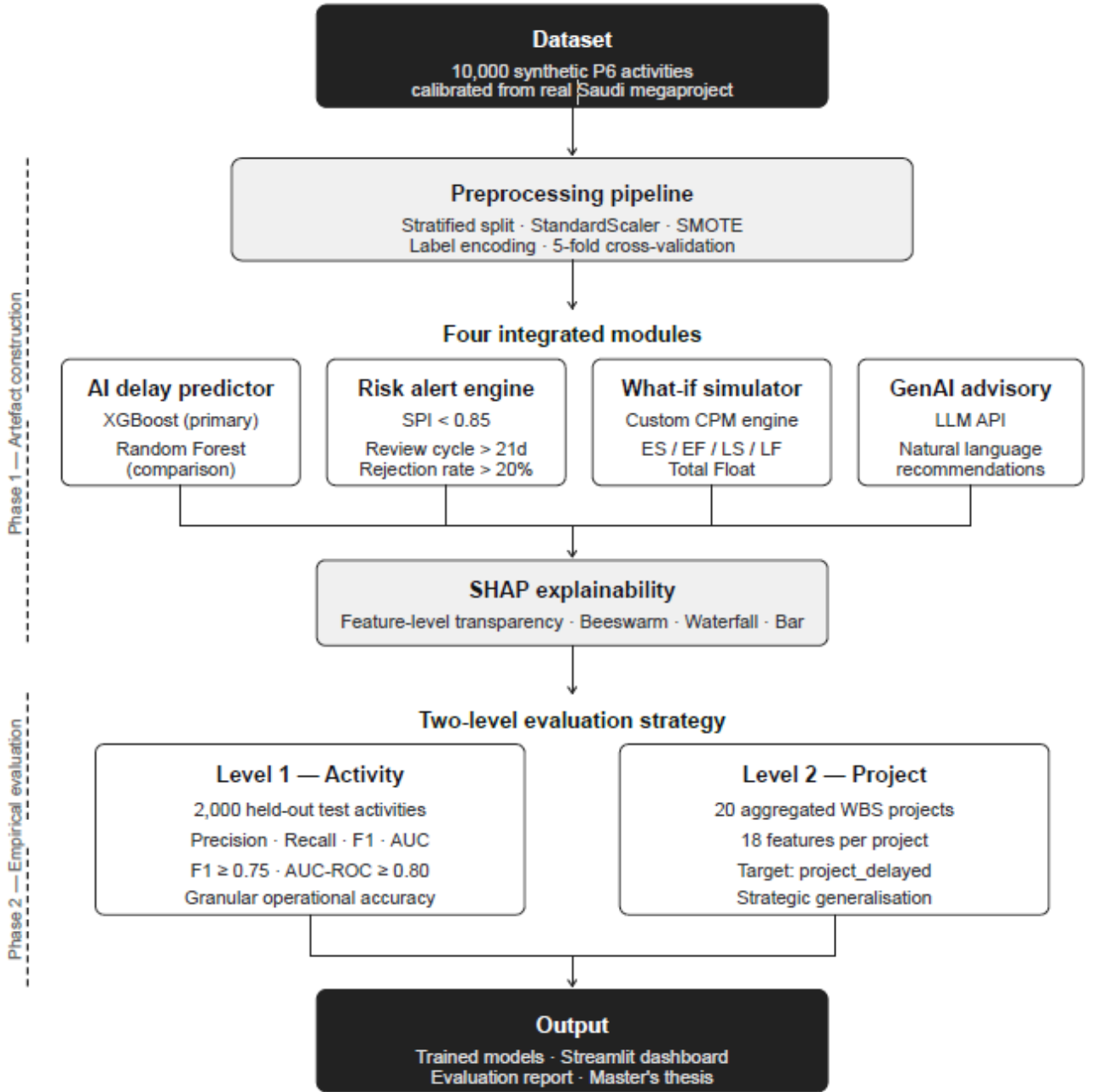


Fig. 1: Proposed AI-powered Digital Twin architecture integrating delay prediction, CPM simulation, risk alerting, SHAP explainability, and generative AI advisory.

activity's delay probability into additive feature contributions. Through this layered interpretability framework, the predictive models are transformed from opaque black box systems into transparent decision support tools that can be interrogated and trusted by project managers.

d. What If CPM Simulator

The What If CPM Simulator implements a custom Critical Path Method (CPM) engine designed to model and analyze project scheduling dynamics. The engine computes key scheduling parameters, including Early Start (ES), Early Finish (EF), Late Start (LS), Late Finish (LF), and Total Float (TF), for each activity within the project network. It

supports real time recalculation in response to modifications in activity durations or predecessor relationships, enabling immediate generation of an updated project completion date and a revised critical path. Additionally, the simulator identifies near critical activities characterized by a total float less than or equal to two days ($TF \leq 2$) and estimates the resulting cost impact based on a configurable daily overhead rate. This functionality enables rapid scenario evaluation and supports proactive schedule optimization and decision making.

e. Proactive Risk Alert Engine

The Proactive Risk Alert Engine functions as a rule based monitoring layer continuously applied to incoming project

data streams. It evaluates key performance indicators against predefined thresholds to detect early signs of potential project risks. Specifically, alerts are triggered when the Schedule Performance Index (SPI) falls below 0.85, when document review cycle durations exceed 21 days, when submittal rejection rates surpass 20%, or when progress deviation drops below -10% relative to the planned baseline. Upon detection of threshold violations, the system generates structured warning notifications that identify the affected activity or project zone, the triggering condition, and recommended immediate mitigation actions. This mechanism facilitates early risk detection and enables timely intervention before issues escalate into critical delays.

f. Generative AI Advisory Chat

The Generative AI Advisory Chat integrates a large language model through an application programming interface (API) to provide context aware decision support via natural language interaction. Project managers can describe emerging challenges using a conversational interface, enabling intuitive engagement with the system. Each query is enriched with relevant contextual project data, including Schedule Performance Index (SPI), Cost Performance Index (CPI), delay probability scores, and current critical path status, before being submitted to the model. Based on this enriched input, the system generates tailored recovery recommendations, schedule acceleration strategies, and resource reallocation proposals aligned with the specific project conditions. This capability transforms the Digital Twin from a passive analytical system into an interactive and intelligent advisory platform that actively supports real time project management decisions.

g. Validation Strategy

To evaluate the effectiveness and practical applicability of the proposed system, a two level validation strategy is adopted, addressing both operational and strategic perspectives. At the activity level, the predictive models are evaluated using a held out test dataset comprising 2,000 activities, with performance assessed through standard classification metrics. At the project level, the trained models are applied to 20 aggregated project records to assess their ability to generalize from activity level patterns to overall project delay outcomes. This dual level validation approach ensures that the system not only achieves strong predictive accuracy at a granular level but also demonstrates meaningful applicability in real world project management scenarios.

h. Dataset Description

The primary training dataset comprises 10,000 synthetic Primavera P6 activity records generated using a domain calibrated data generation pipeline. The statistical parameters governing data generation were derived from a real Primavera P6 export of a Saudi Arabian construction megaproject comprising 28,521 activities over 3,877 calendar days, exhibiting a physical completion percentage of 86.92%, a schedule completion percentage of 95.40%,

and a performance completion percentage of 95.65%.

All project identifying information was excluded from the research; only aggregate statistical properties were utilized to calibrate the synthetic generation process, ensuring compliance with confidentiality requirements. The dataset contains no missing values across all 25 fields.

1. Feature Dictionary

The dataset captures 25 features per activity record as summarized in Table 2.

TABLE 2: PRIMARY DATASET — FEATURE DICTIONARY

Feature	Type	Description
activity_id	String	Unique activity code
activity_name	String	Descriptive activity name
wbs_code	String	Work Breakdown Structure code
activity_type	Categorical	Type of construction work
zone	Categorical	Spatial zone
level	Categorical	Floor level
area	Categorical	Sub area
primary_resource	Categorical	Assigned resource
planned_duration_days	Integer	Planned duration
actual_duration_days	Integer	Actual duration
delay_days	Integer	Delay duration
planned_start	Date	Planned start date
planned_finish	Date	Planned finish date
actual_finish	Date	Actual finish date
physical_pct_complete	Float	Physical progress
duration_pct_complete	Float	Time progress
schedule_pct_complete	Float	Schedule completion
performance_pct_complete	Float	Combined performance
spi	Float	Schedule Performance Index
cpi	Float	Cost Performance Index
bl_labor_units	Float	Planned labor hours
bl_labor_cost_sar	Float	Labor cost (SAR)
revised_weight	Float	Activity weight
delayed	Binary	Delay label
delay_magnitude_days	Integer	Delay magnitude

The binary target variable *delayed* is defined as 1 when an activity exceeds its planned finish date and 0 otherwise, yielding a delay rate of 45.8% (4,583 delayed activities). Zone-4 activities and commissioning phases exhibit the highest delay propensity.

Exploratory analysis indicates that Schedule Performance Index (SPI) is the most strongly correlated feature with delay occurrence, followed by physical completion percentage, schedule completion percentage, and Cost

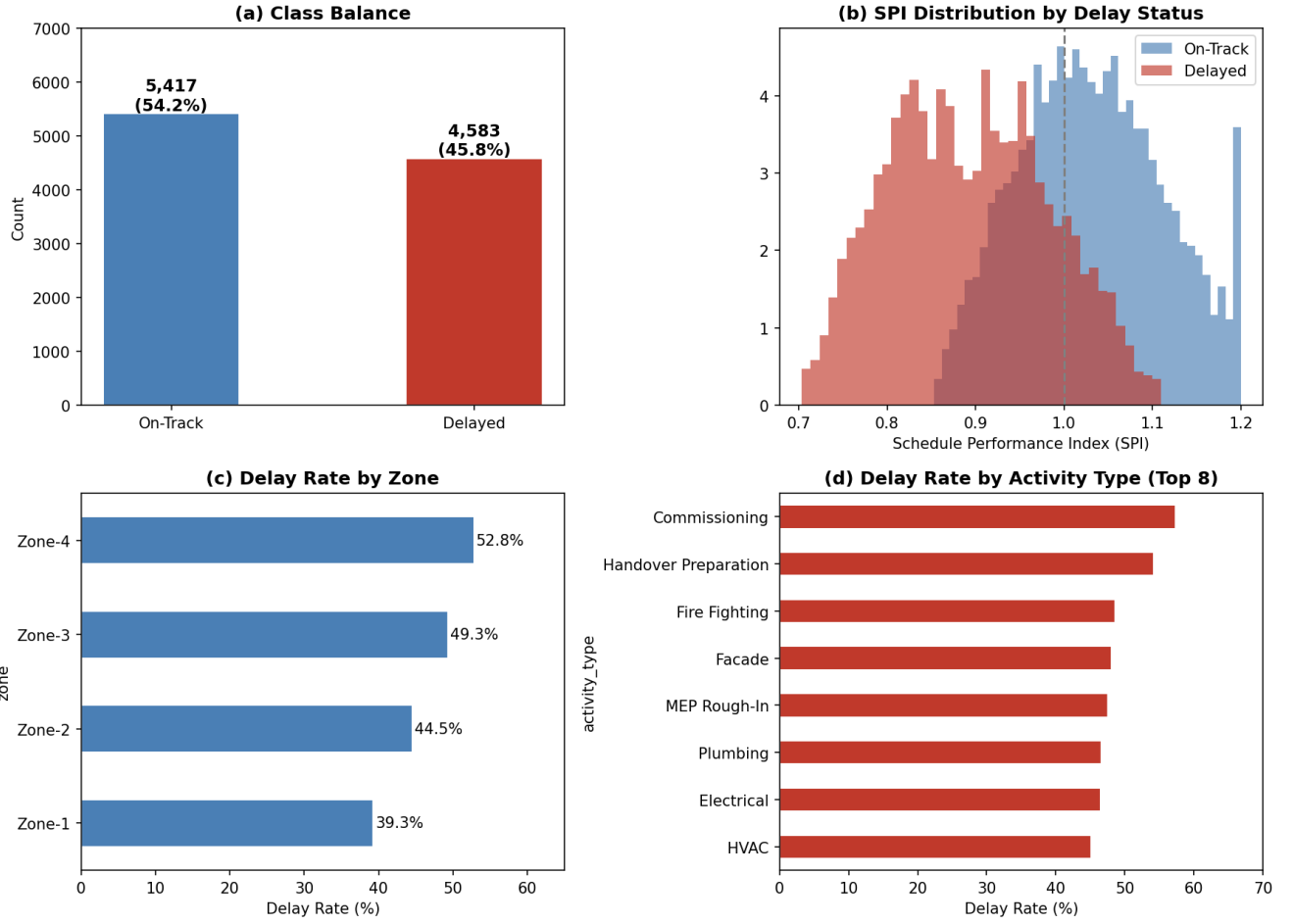


Fig. 2: Exploratory data analysis of the synthetic P6 dataset ($n = 10,000$). (a) Class distribution showing 45.8% delayed activities. (b) SPI distribution by delay status, where delayed activities exhibit consistently lower SPI values. (c) Delay rate by spatial zone, with Zone-4 showing the highest delay rate. (d) Delay rate by activity type, with Commissioning and Handover Preparation exhibiting the highest delay propensity.

Performance Index (CPI). No missing values were observed across all features.

2. Project Level Dataset

A secondary dataset was derived by aggregating the activity level data into 20 project level records based on Work Breakdown Structure (WBS) codes. Each record includes 18 aggregated features such as average SPI, average CPI, total delay days, activity count, delayed activity count, rejection rate, review cycle duration, and resource utilization.

The target variable *project_delayed* is defined as 1 when more than 15% of a project's activities are delayed, enabling evaluation of model generalization at a strategic level.

3. Data Preprocessing and Evaluation Pipeline

All data undergoes a structured data preparation pipeline prior to model training. The process begins with a stratified train-test split (80% training, 20% testing) to preserve class distribution and prevent evaluation bias. Feature scaling is then applied using StandardScaler fitted on the training set and applied to the test set to avoid data leakage. To address class imbalance, SMOTE oversampling ($k = 5$) is applied exclusively to the training data.

Categorical variables, including activity type, zone, level, WBS code, and resource, are transformed using label encoding to ensure compatibility with machine learning algorithms. Missing value handling is implemented using a `fillna(0)` safeguard to ensure robustness, although no missing values are present in the dataset.

TABLE 3: DATA PREPROCESSING PIPELINE

Step	Procedure	Rationale
1	Stratified split (80% training, 20% test)	Prevents class imbalance bias
2	Feature scaling using StandardScaler	Prevents data leakage
3	SMOTE oversampling ($k=5$)	Addresses class imbalance
4	Label encoding for categorical variables	Converts features to numerical format
5	Missing value handling (<code>fillna(0)</code>)	Ensures robustness
6	Stratified 5-fold cross-validation	Evaluates generalization performance

IV. CONCLUSIONS

The foundational phases of this research have been completed, encompassing problem definition, literature review, methodology design, and dataset preparation. Three persistent operational failures motivating the study — slow manual reporting, fragmented information systems, and intuition based decision making — have been grounded in both the literature and direct professional experience. Five critical research gaps have been identified, and a Design Science Research methodology has been adopted to address them through four integrated intelligent modules. A synthetic Primavera P6 dataset of 10,000 activities calibrated from a real Saudi Arabian construction megaproject has been generated, and the preprocessing pipeline has been fully specified. The work completed establishes a solid foundation for Phase 2, which will focus on model training, SHAP explainability analysis, Streamlit dashboard development, and Generative AI Advisory Chat integration.

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